



How to update Kibo RPC Game API and Dependencies

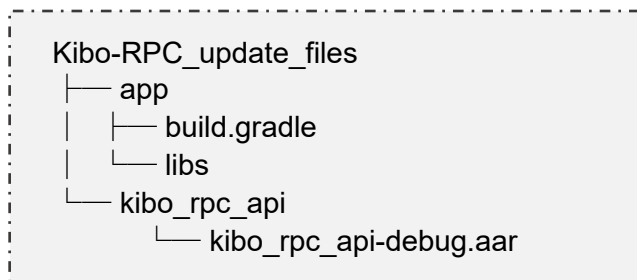
1. Overview

- Kibo RPC API Updates
 - We've updated the API so that when run in the simulator, it now returns images with noise.
- Game APK Dependencies Changes
 - Some of the rosjava dependency libraries have become unavailable publicly. We've added instructions on how to reference the physical files directly.

2. Update Files

Click <https://jaxa.krpc.jp/download> > [Kibo-RPC UPDATE FILES] to download and unzip the zip file.

The ZIP file includes the following structure:



3. Instruction

Follow this instruction and apply new API library and Some of the rosjava dependency libraries to your program.

1. Launch Android Studio.
2. Right-click the " kibo_rpc_api" and select [Show in Files].

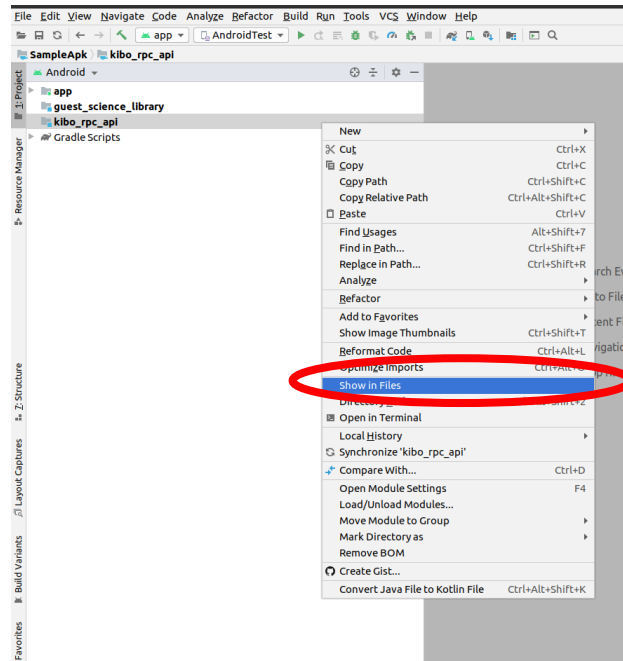


Figure 1 Select “Show in Files”

3. Please replace and overwrite the existing “kibo_rpc_api-debug.aar” file in the opened folder with the latest version.

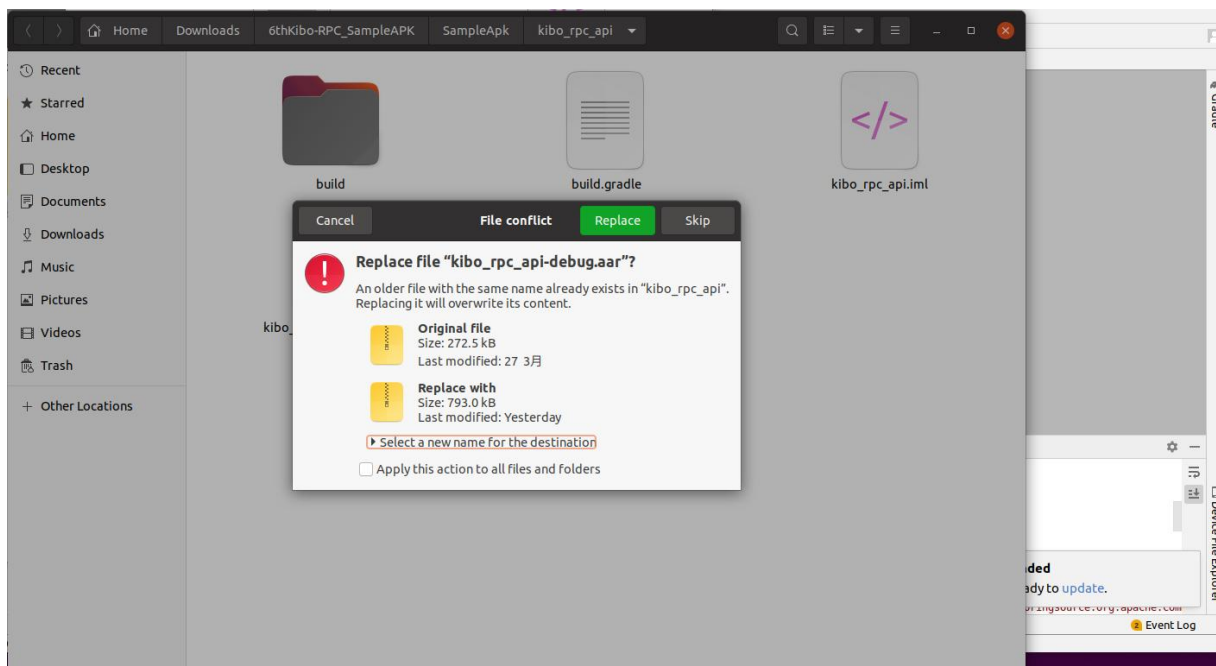


Figure 2 Overwrite “kibo_rpc_api-debug.aar”

4. Next, let's set up the rosjava dependencies. Begin by showing the Project view in Android Studio.

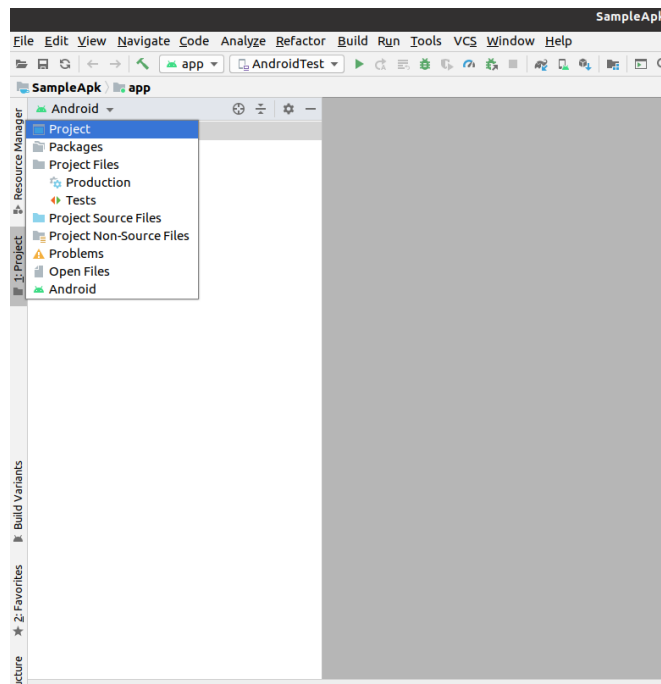


Figure 3 Select “Project”

5. right-click on the app directory, select [New]>[Directory], and create a folder named libs.

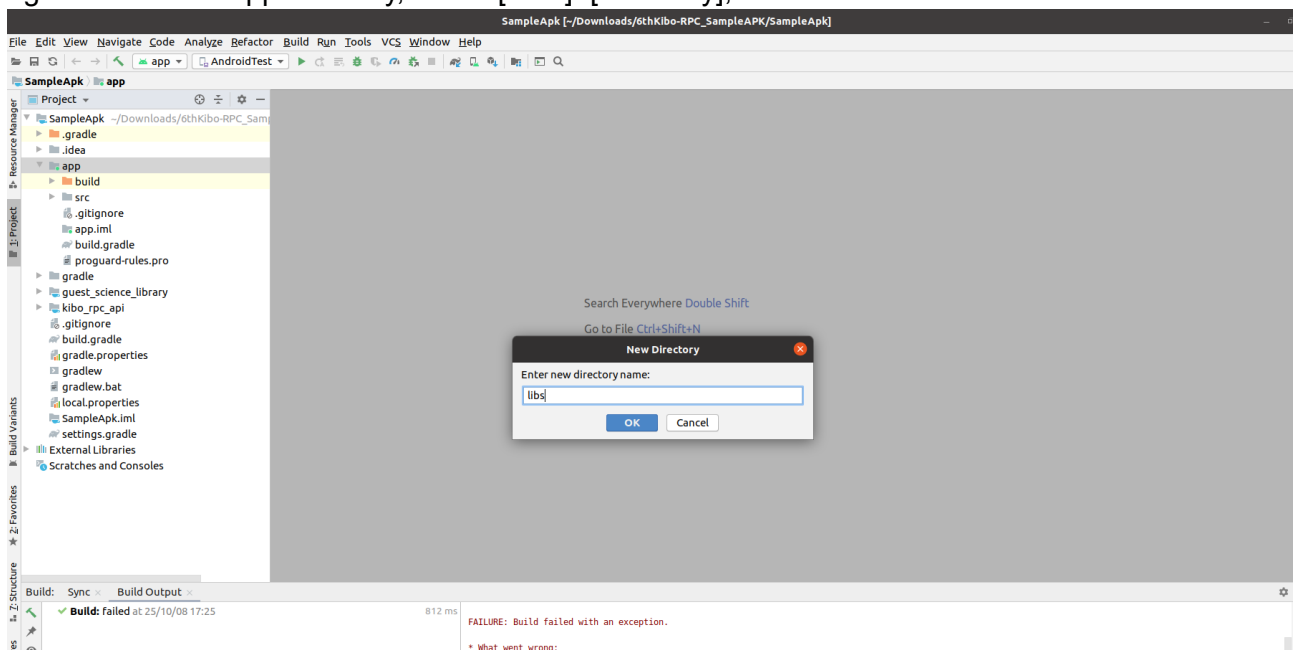


Figure 4 Create “libs”



- Copy the JAR files located under app/libs in the extracted folder into your project's libs directory.

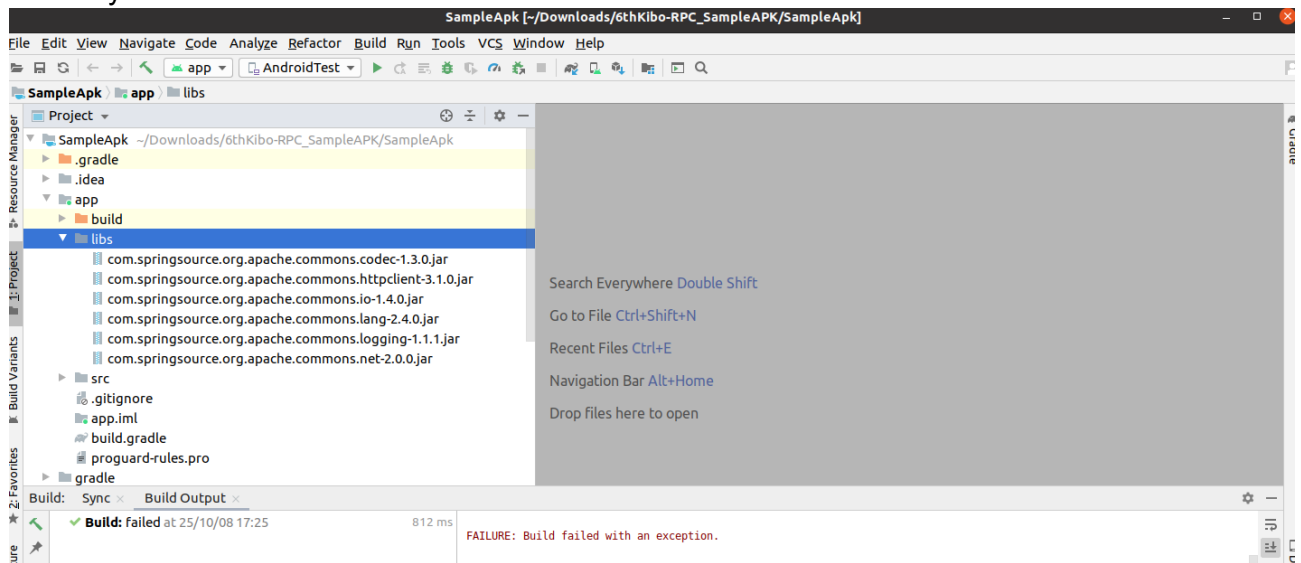


Figure 5 copy jar files to “libs”

- Add the necessary settings to your custom program's **app>build.gradle** file.
You can refer to the sample file located in Kibo-RPC_update_packages>app>build.gradle.

If you've added libraries to the provided APK, please follow the steps below.
If there are no changes from the provided APK, simply overwriting the file is fine.

Note: Please note that there are other files named **build.gradle** in different directories.
Make sure you're editing the one under the **app** directory.

- Please add the following **configurations** at line 39

```
configurations {  
    all*.exclude group: 'org.apache.commons', module: 'com.springsource.org.apache.commons.httpclient'  
    all*.exclude group: 'org.apache.commons', module: 'com.springsource.org.apache.commons.codec'  
    all*.exclude group: 'org.apache.commons', module: 'com.springsource.org.apache.commons.io'  
    all*.exclude group: 'org.apache.commons', module: 'com.springsource.org.apache.commons.lang'  
    all*.exclude group: 'org.apache.commons', module: 'com.springsource.org.apache.commons.logging'  
    all*.exclude group: 'org.apache.commons', module: 'com.springsource.org.apache.commons.net'  
}
```

- Include the following **implementation** under dependencies.

```
implementation files('libs/com.springsource.org.apache.commons.codec-1.3.0.jar')  
implementation files('libs/com.springsource.org.apache.commons.io-1.4.0.jar')  
implementation files('libs/com.springsource.org.apache.commons.lang-2.4.0.jar')  
implementation files('libs/com.springsource.org.apache.commons.httpclient-3.1.0.jar')  
implementation files('libs/com.springsource.org.apache.commons.logging-1.1.1.jar')  
implementation files('libs/com.springsource.org.apache.commons.net-2.0.0.jar')
```

8. Please make sure that your app>build.gradle file looks like the following.

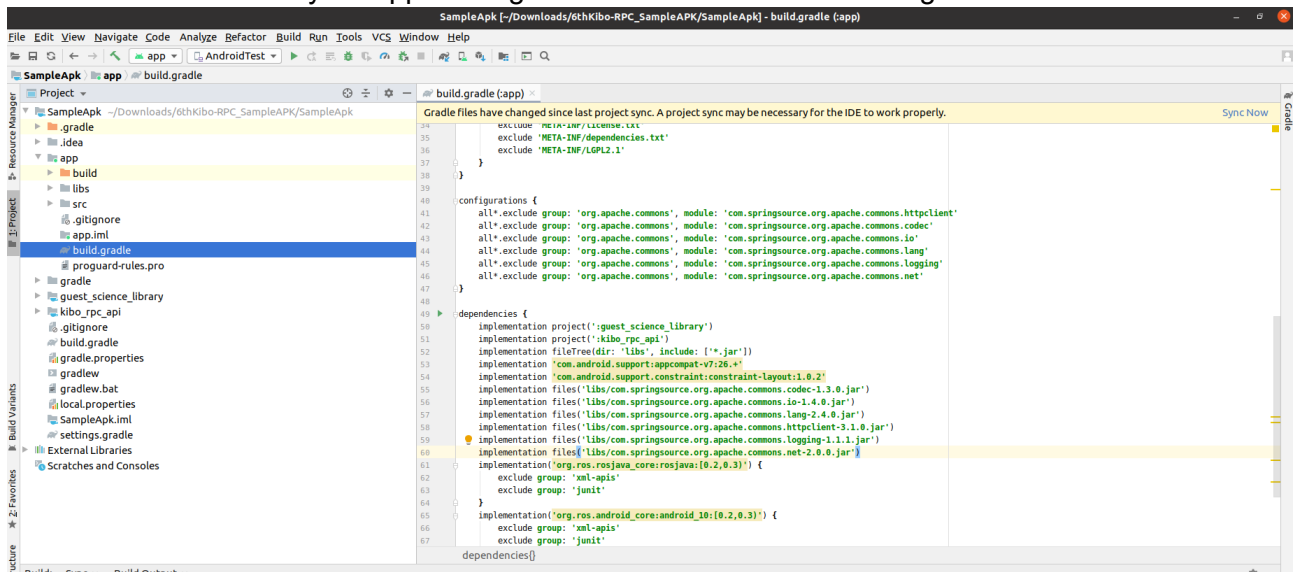
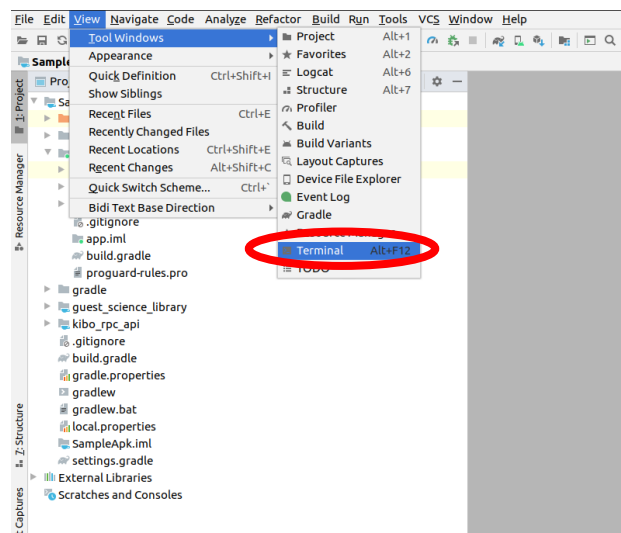


Figure 6 Updated “build.gradle”

9. From the menu bar, go to [View]>[Tool Windows]>[Terminal], and run the following command.



☒ 7 Select “Terminal”

- For Windows:

```
gradlew cleanBuildCache
```

- For Ubuntu:

```
./gradlew cleanBuildCache
```

10. Rebuild the APK (see Section 3.3 of the Programming Manual).
If the build is completed successfully, the library replacement has been applied correctly.
11. Confirm that the latest version of the Kibo-RPC API has been properly applied.
Run the rebuilt APK in the simulator.



If the Android Emulator log (adb.log) contains the line

KiboRpcApi: [AARVersion] AAR version 6.1.0, then the latest version has been successfully applied.

4. List of Changes

All changes to paragraphs, tables, and figures in this document are shown below;

Release Date	Revision	Paragraph(s)	Rationale
Oct. 15 th , 2025	1.0	All	-